

The answer key for the **Guts** round is below.

p	Answer	p	Answer	p	Answer
1	0	2	2025	3	24
4	972	5	703	6	717
7	37	8	$\frac{\sqrt{7}}{4} + 1$	9	60
10	21	11	$\frac{11\sqrt{3}}{3}$	12	$\frac{7}{60}$
13	4975	14	$\sqrt{6}$	15	3168
16	112	17	256	18	$-\frac{\sqrt{3}}{7}$
19	5760	20	$\frac{6}{325}$	21	$\frac{14}{55}$
22	$\frac{2+\sqrt{2}+\sqrt{6}}{4}$	23	24100	24	$\frac{16\sqrt{3}}{243}$
25	286	26	$\sqrt{61}$	27	10

Problem 1. [5] Real numbers $x_1, x_2, \dots, x_{1434}$ satisfy $|x_n - x_{n+1}| = n^2$ for all integers $1 \leq n \leq 1433$. What is the sum of all possible values of $x_1 - x_{1434}$?

Proposed by: Jefferson Zhou

Answer: 0

Solution: Observe that $x_1 - x_{1434} = (x_1 - x_2) + (x_2 - x_3) + \dots + (x_{1433} - x_{1434}) = \pm 1^2 \pm 2^2 \pm \dots \pm 1433^2$. Thus, for any value a of $x_1 - x_{1434}$, $-a$ is also obtainable, and thus the answer is 0.

Problem 2. [5] A right triangle has hypotenuse of length 75 and a leg of length 45. A square has the same perimeter as the triangle. Find the area of the square.

Proposed by: Taison Scofield

Answer: 2025

Solution: By the Pythagorean Theorem, the length of the other leg is $\sqrt{75^2 - 45^2} = \sqrt{3600} = 60$. Hence, the perimeter of the triangle is $45 + 60 + 75 = 180$. The perimeter of a square with side length s is $4s$, so we have $4s = 180 \implies s = 45$. Therefore, the area is $s^2 = \span style="border: 1px solid black; padding: 0 5px;">2025.$

Problem 3. [5] Let the 6-digit number $14A34B$ be divisible by 72. Find $10A + B$.

Proposed by: Taison Scofield

Answer: 24

Solution: Divisibility by 72 implies divisibility by 8 and 9. For a number to be divisible by 8, the last three digits must be divisible by 8. We have $340 + B \equiv 0 \pmod{8} \implies B \equiv 4 \pmod{8}$. Hence, the only digit B that satisfies this is $B = 4$. For a number to be divisible by 9, the sum of the digits must be divisible by 9. We can set up the equation $1 + 4 + A + 3 + 4 + 4 \equiv 0 \pmod{9} \implies A \equiv 2 \pmod{9}$. Thus, $A = 2$, so our answer is 24.

Problem 4. [6] Pentagonal prisms M and N have the same height and have bases that are regular pentagons. The length of a side of the base of M is 6 times that of N . If the combined volumes is 999, find the volume of M .

Proposed by: Taison Scofield

Answer: 972

Solution: Since the height is the same and the bases are regular polygons, the volume of each prism is proportional to the area of its base. The area of a regular polygon scales with the square of the side length, so the volume ratio is: $\frac{V_M}{V_N} = \left(\frac{6s}{s}\right)^2 = 36$. Hence, we have $V_M = 36V_N$ and $V_M + V_N = 999$. Solving the system yields:

$$\begin{aligned} 36V_N + V_N &= 999 \\ 37V_N &= 999 \\ V_N &= 27 \\ V_M &= 36 \cdot 27 = \span style="border: 1px solid black; padding: 0 5px;">972 \end{aligned}$$

Problem 5. [6] How many nonempty sets of the first 10 positive integers contain at most 2 prime numbers?

Proposed by: Taison Scofield

Answer: 703

Solution: First, let's just consider the possible prime numbers 2, 3, 5, and 7. Out of the 16 possible sets with these prime numbers, $\binom{4}{0} + \binom{4}{1} + \binom{4}{2} = 11$ contain 2 prime numbers or less. It doesn't matter if the other 6 numbers are included or not since they're not prime.

Thus, we get $11 \cdot 2^6 = 704$ sets, and subtracting the empty set, our final answer is 703.

Problem 6. [6] Given that there is exactly one solution to the system of equations

$$\begin{aligned}y &= 3x^2 + 24x + c \\ y &= 669,\end{aligned}$$

find c .

Proposed by: Taison Scofield

Answer: 717

Solution: Note that $y = 3x^2 + 24x + c$ is a parabola that opens upwards and $y = 669$ is a horizontal line. That means the one solution must occur at the parabola's vertex because of symmetry, so the vertex must have a y -coordinate of 669.

$$\begin{aligned}x &= -\frac{b}{2a} = -\frac{24}{2 \cdot 3} = -4 \\ 669 &= 3(-4)^2 + 24(-4) + c \\ 669 &= 48 - 96 + c \\ 669 &= -48 + c \\ c &= 669 + 48 = \span style="border: 1px solid black; padding: 0 5px;">717\end{aligned}$$

Problem 7. [7] LeBron James enjoys playing basketball. For every 2-point shot LeBron takes, he has a $\frac{1}{2}$ chance of scoring and getting 2 points. For every 3-point shot LeBron takes, he has a $\frac{1}{3}$ chance of scoring and getting 3 points. LeBron has a score sequence that starts empty. For every 2-point shot he scores, he appends a 2 to the end of his score sequence, and for every 3-point shot he takes, he appends a 3 to the end of his score sequence. If LeBron ended up scoring 16 points, how many possible score sequences could LeBron have?

Proposed by: Zack Berchenko

Answer: 37

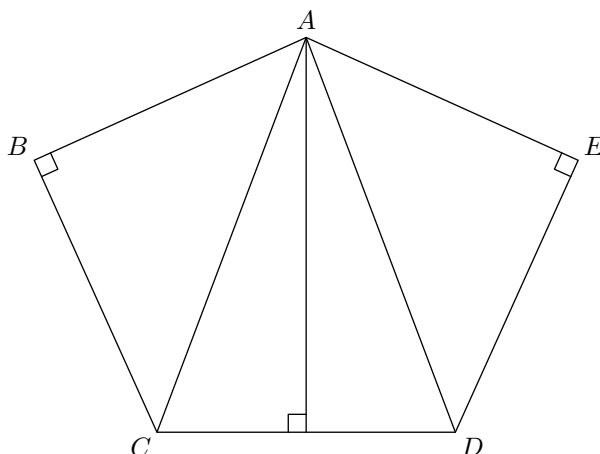
Solution: Case 1: Two 2-point shots, and four 3-point shots: $\binom{6}{2} = 15$. Case 2: Five 2-point shots, and two 3-point shots: $\binom{7}{2} = 21$. Case 3: Eight 2-point shots, zero 3-point shots: 1 Total is $15 + 21 + 1 = 37$.

Problem 8. [7] Equilateral pentagon $ABCDE$ has side length 1, and $\angle E = \angle B = 90^\circ$. Compute the area of $ABCDE$.

Proposed by: Ryan Niu

Answer: $\boxed{\frac{\sqrt{7}}{4} + 1}$

Solution: Note that by drawing in segments AC and AD , we can clearly see the pentagon is made up of 2 right isosceles triangles, and one central isosceles triangle. By the Pythagorean theorem we get $AC = AD = \sqrt{2}$, and we can apply again to calculate the height of $\triangle ACD$ to be $\frac{\sqrt{7}}{2}$. Using all of this, we get the area to be $\boxed{\frac{\sqrt{7}}{4} + 1}$.



Problem 9. [7] Let $f(n)$ represent the number of positive integer divisors of n . What is the minimum value of n that satisfies $f(f(f(n))) = 4$?

Proposed by: Taison Scofield

Answer: $\boxed{60}$

Solution: We work from the outside in. First, $f(x) = 4$ when $x = p^3$ or $x = pq$ for distinct primes p and q . The smallest such x is 6 and the second smallest possible value of x is 8, so $f(f(n)) = 6$ or 8.

Now, $f(k) = 6$ when k has divisor count 6. The same is true with $k = 8$. Possible forms include:

$$\begin{aligned} p^5 & \text{ (divisors: 6),} \\ p^2q & \text{ (divisors: } (2+1)(1+1) = 6\text{),} \\ pqr & \text{ (all exponents 1: } (1+1)^3 = 8\text{).} \end{aligned}$$

The smallest such k is $12 = 2^2 \cdot 3$, so $f(n) = 12$.

We now find the smallest n such that $f(n) = 12$. We want to minimize n under the constraint that the product of one more than its exponents equals 12. The factorization $12 = 3 \cdot 2 \cdot 2$ corresponds to exponents 2, 1, 1, giving

$$n = 2^2 \cdot 3 \cdot 5 = 60.$$

Check:

$$f(60) = (2+1)(1+1)(1+1) = 3 \cdot 2 \cdot 2 = 12$$

$$\begin{aligned} f(f(60)) &= f(12) = 6 \\ f(f(f(60))) &= f(6) = 4 \end{aligned}$$

Thus, the minimum such n is $\boxed{60}$.

Problem 10. [9] Let $x + \frac{2}{x} = 5$. Determine $x^2 + \frac{4}{x^2}$.

Proposed by: Zack Berchenko

Answer: $\boxed{21}$

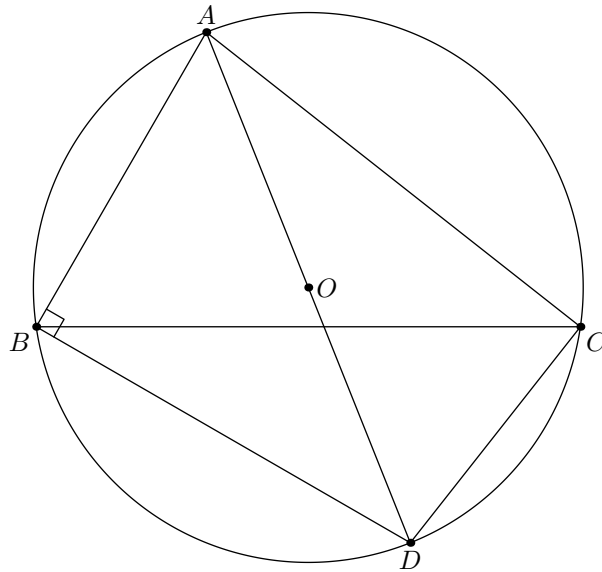
Solution: Square the first equation to obtain $x^2 + 4 + \frac{4}{x^2} = 25$ or $x^2 + \frac{4}{x^2} = \boxed{21}$.

Problem 11. [9] Let $\triangle ABC$ have side lengths $AB = 5$, $AC = 7$, and $BC = 8$ with circumcenter O , and let D be the intersection of AO with its circumcircle. Find BD .

Proposed by: Angie Huang

Answer: $\boxed{\frac{11\sqrt{3}}{3}}$

Solution:



Note: $[ABC]$ denotes the area of triangle ABC .

By Heron's, $[ABC] = \sqrt{10 \cdot 2 \cdot 3 \cdot 5} = 10\sqrt{3}$, so $\frac{5 \cdot 7 \cdot 8}{4R} = [ABC] = 10\sqrt{3}$, where R is the circumradius. Now, $R = \frac{7}{\sqrt{3}}$.

Observe that AD is a diameter of the circumcircle of ABC , so $\angle ABD = 90^\circ$, and $AD = 2R = \frac{14}{\sqrt{3}}$. Now, by the Pythagorean Theorem, $BD = \sqrt{AD^2 - AB^2} = \sqrt{\frac{196}{3} - 25} = \sqrt{\frac{121}{3}} = \boxed{\frac{11\sqrt{3}}{3}}$.

Problem 12. [9] Xook tells Xonk and Xoink to write a permutation of the word "STORY" uniformly at random. Xonk does this, but Xoink mishears the request and writes down a random permutation of the word "DIRTY". What is the probability that Xonk's and

Xoink's permutations match in at least 2 spots? For example, if Xonk wrote "STORY" and Xoink wrote "DIRTY", then their permutations would match in 1 spot, namely the final Y.

Proposed by: Jefferson Zhou

Answer: $\boxed{\frac{7}{60}}$

Solution: Observe that the only letters common to both STORY and DIRTY are the R, T, and Y.

WLOG, let Xonk's permutation be STORY (only the relative order of the letters matters, so this is valid.) There are $3! = 6$ ways for the R and the T to match, there are $3! = 6$ ways for the T and Y to match, and there are $3! = 6$ ways for the R and Y to match. However, we counted the cases where the R , T , and Y match 3 times, but we only want to count them one time. There are $2! = 2$ ways for the R , T , and Y to match, so we need to subtract by 4. Now, there are $6 + 6 + 6 - 4 = 14$ possible ways for the permutations to match in at least two spots. Also, there are $5! = 120$ possible permutations of the word DIRTY, so the answer is $\frac{14}{120} = \boxed{\frac{7}{60}}$.

Problem 13. [10] How many positive integers n less than 10^4 have a digit a when written in base 10 such that a^2 and n have the same units digit?

Proposed by: Jefferson Zhou

Answer: $\boxed{4975}$

Solution: We need to consider the quadratic residues modulo 10: they are 0, 1, 4, 5, 6, 9. Out of these, $0^2 \equiv 0 \pmod{10}$, $1^2 \equiv 1 \pmod{10}$, $5^2 \equiv 5 \pmod{10}$, and $6^2 \equiv 6 \pmod{10}$ so any positive integers ending in 0, 1, 5, 6 automatically satisfy the conditions. There are $3 + 999 \cdot 4 = 3999$ positive integers less than 10^4 which end in these digits. We now consider the case where the number ends in 4 or 9.

Observe that $2^2, 8^2 \equiv 4 \pmod{10}$ and $3^2, 7^2 \equiv 9 \pmod{10}$, so any positive integer ending in 4 with this condition must contain the digits 2 or 8. Also, observe that the cases will have the same number of solutions, so we just focus on the case in which n ends in 4.

If the number is 2 digits long, then there are 2 possible numbers. If the number is 3 digits long, then we can use complementary to yield that there are $90 - 56 = 34$ possible numbers, and if the number is 4 digits long, then we use complementary to yield that there are $900 - 448 = 452$. Thus, there are $2 + 34 + 452 = 488$ possible numbers which end in 4.

Now, the answer is $3999 + 488 \cdot 2 = \boxed{4975}$.

Problem 14. [10] The real numbers x and y satisfy the system of equations

$$\begin{aligned} x^2 + 2xy + y^2 &= \frac{10}{x} + \frac{10}{y} \\ x^3 + y^3 &= 34. \end{aligned}$$

Compute $|x - y|$.

Proposed by: Jefferson Zhou

Answer: $\boxed{\sqrt{6}}$

Solution: The first equation simplifies to $(x + y)^2 = \frac{10(x + y)}{xy}$. Observe that if $x + y = 0$, then $x = -y$, so $x^3 + y^3 = 0$, a contradiction. Thus, $x + y \neq 0$, so $x + y = \frac{10}{xy}$, and this simplifies to $x^2y + xy^2 = 10$. This looks a lot like the middle terms of the expansion of $(x + y)^3$, so we multiply it by 3 and add it to the second equation, yielding $(x + y)^3 = x^3 + 3x^2y + 3xy^2 + y^3 = 30 + 34 = 64$. Since x, y are real, we have $x + y = 4$, so now $xy = \frac{10}{x + y} = \frac{5}{2}$. Thus, x, y are the roots of $z^2 - 4z + \frac{5}{2} = 0$, so $x, y = \frac{4 \pm \sqrt{6}}{2}$. Now, $|x - y| = \left| \frac{4 + \sqrt{6}}{2} - \frac{4 - \sqrt{6}}{2} \right| = \boxed{\sqrt{6}}$.
 Alternatively, $(x - y)^2 = (x + y)^2 - 4xy = 16 - 10 = 6$, so $|x - y| = \sqrt{6}$.

Problem 15. [10] Consider a 23×23 grid of unit squares. The single square located exactly at the center of the grid is removed.

Find the total number of squares (of all sizes, sides parallel to the grid) contained entirely within the grid that do not include the removed center square.

Proposed by: Taison Scofield

Answer: $\boxed{3168}$

Solution: The total number of squares in a complete $N \times N$ grid is $S_{\text{total}} = \sum_{k=1}^N (N - k + 1)^2 = \sum_{j=1}^N j^2 = \frac{N(N+1)(2N+1)}{6}$.

Substituting $N = 23$: $S_{\text{total}} = \frac{23 \cdot 24 \cdot 47}{6} = 4324$.

Next, count the number of squares that contain the removed center cell at $(n+1, n+1) = (12, 12)$.

For $1 \leq k \leq n + 1 = 12$, the number of $k \times k$ squares containing the center is k^2 .

For $12 < k \leq 23$, every such $k \times k$ square contains the center, so the count is $(N - k + 1)^2$.

Hence, the total number of invalid squares is $S_{\text{invalid}} = \sum_{k=1}^{12} k^2 + \sum_{k=13}^{23} (23 - k + 1)^2$.

Calculate each sum: $\sum_{k=1}^{12} k^2 = \frac{12 \cdot 13 \cdot 25}{6} = 650$, and by substituting $j = 23 - k + 1$, the second sum becomes $\sum_{j=1}^{11} j^2 = \frac{11 \cdot 12 \cdot 23}{6} = 506$.

Thus, $S_{\text{invalid}} = 650 + 506 = 1156$.

Finally, the number of valid squares that do not include the removed center cell is $S_{\text{valid}} = S_{\text{total}} - S_{\text{invalid}} = 4324 - 1156 = \boxed{3168}$.

Problem 16. [12] Let S be the set of positive integers such that c is in S if and only if there exist positive integers a, b such that $c = \gcd(a^b, b^a)$. Find the sum of the five smallest numbers in S .

Proposed by: Samuel Kretzschmar

Answer: $\boxed{112}$

Solution: We show that if prime $p \mid c$, then $p \mid v_p(c)$. We see $p \mid c$ implies $p \mid a$ and $p \mid b$. As $v_p(c) = \min(v_p(a^b), v_p(b^a)) = \min(bv_p(a), av_p(b))$ and $p \mid bv_p(a), av_p(b)$, we have $p \mid v_p(c)$ as desired. Noting this, if prime $p \mid c$ and $p > 3$, then $c \geq 5^5 = 3125$. If not, we only consider $p = 2, 3$ instead, so $c = 2^{2m}3^{3n}$ for nonnegative integers m, n . Not all integers of this form are in S ; however, the first five ($1 = \gcd(1^1, 1^1)$, $2^2 = 4 = \gcd(2^2, 2^2)$, $2^4 = 16 = \gcd(2^4, 4^2)$, $3^3 = 27 = \gcd(3^3, 3^3)$, $2^6 = 64 = \gcd(8^2, 2^8)$) all are with these constructions. These sum to $1 + 4 + 16 + 27 + 64 = 112$.

Problem 17. [12] How many permutations $f(x)$ of the set $\{1, 2, 3, 4, 5, 6\}$ satisfy $f(f(f(f(x)))) = x$?

Proposed by: Angie Huang

Answer: $\boxed{256}$

Solution: Let $S = \{1, 2, 3, 4, 5, 6\}$.

We first do the case where there is one cycle of length 4. The other two elements of S can either be in a cycle of 2, or they can be sent to themselves. In the former case, there are $\binom{6}{2}$ ways to choose which elements are in the 4-cycle, and there are $3!$ ways to decide the order of the cycle, so there are $15 \cdot 6 = 90$ possibilities. In the latter case, the logic is identical, and there are also 90 possibilities.

We now do the case where there are no cycles of length 4. If there are 3 cycles of length 2, then there are $\frac{\binom{6}{2}\binom{4}{2}\binom{2}{2}}{3!} = 15$ possibilities, if there are 2 cycles of length 2, then there are $\frac{\binom{6}{2}\binom{4}{2}}{2!} = 45$ possibilities, if there is 1 cycle of length 2, then there are $\binom{6}{2} = 15$ possibilities, and if there are no cycles of length 2, then there is only one possibility, namely the identity function.

Thus, in total, there are $90 + 90 + 15 + 45 + 15 + 1 = \boxed{256}$ possible permutations.

Problem 18. [12] Let $S = \sum_{k=1}^{\infty} \frac{\sin\left(\frac{4\pi k}{3}\right)}{2^k} = \frac{\sin\left(\frac{4\pi}{3}\right)}{2^1} + \frac{\sin\left(\frac{8\pi}{3}\right)}{2^2} + \dots$. Compute the value of S .

Proposed by: Taison Scofield

Answer: $\boxed{-\frac{\sqrt{3}}{7}}$

Solution: Since $\sin\left(\frac{4\pi k}{3}\right)$ repeats every 3 terms, split S into sums over $k \equiv 1, 2, 0 \pmod{3}$.

Calculate the sine values: - For $k = 3m + 1$, $\sin\left(\frac{4\pi}{3}\right) = -\frac{\sqrt{3}}{2}$, - For $k = 3m + 2$, $\sin\left(\frac{8\pi}{3}\right) = \sin\left(\frac{2\pi}{3}\right) = \frac{\sqrt{3}}{2}$, - For $k = 3m + 3$, sine is 0.

Thus,

$$S = -\frac{\sqrt{3}}{2} \cdot \sum_{m=0}^{\infty} \frac{1}{2^{3m+1}} + \frac{\sqrt{3}}{2} \cdot \sum_{m=0}^{\infty} \frac{1}{2^{3m+2}}.$$

Each sum is geometric:

$$\sum_{m=0}^{\infty} \frac{1}{2^{3m+1}} = \frac{\frac{1}{2}}{1 - \frac{1}{8}} = \frac{4}{7}, \quad \sum_{m=0}^{\infty} \frac{1}{2^{3m+2}} = \frac{\frac{1}{4}}{1 - \frac{1}{8}} = \frac{2}{7}.$$

Putting everything together, we get

$$S = -\frac{\sqrt{3}}{2} \cdot \frac{4}{7} + \frac{\sqrt{3}}{2} \cdot \frac{2}{7} = \frac{\sqrt{3}}{2} \cdot \left(-\frac{4}{7} + \frac{2}{7}\right) = \boxed{-\frac{\sqrt{3}}{7}}.$$

Problem 19. [14] Call an integer n swell if $\frac{n^n + n!}{n+1}$ is also an integer. Find the product of the first six positive swell integers.

Proposed by: Samuel Kretzschmar

Answer: 5760

Solution: If $n+1$ is composite, then $n+1 \mid n!$ if $n+1 \neq 4$, so $(n^n + n!) \equiv (-1)^n + 0 \not\equiv 0 \pmod{n+1}$ and therefore n is not swell. We note $n = 3$ with $n+1 = 4$ is also not swell, as $\frac{3^3 + 3!}{3+1} = \frac{33}{4}$. If $n+1$ is an odd prime (n even), then using Wilson's, $n^n + n! \equiv (-1)^n + (-1) \equiv 0 \pmod{n+1}$ and so n is swell. Noting $n = 1$ is also swell, we therefore get the first six swell integers as $1, 3-1=2, 5-1=4, 7-1=6, 11-1=10, 13-1=12$ with product 5760.

Problem 20. [14] Thinking that AM-GM applies to negative real numbers, Jeff derives the (incorrect) inequality $27 - a - b \geq 9\sqrt[3]{ab}$ for all a, b . If we uniformly at random chose two not necessarily distinct integers a and b between 0 and 64, inclusive, what is the probability that they satisfy this inequality? You may find it helpful to reference the table below, which lists the number $\sqrt[3]{2}$ to $\sqrt[3]{7}$ to the nearest thousandth.

$\sqrt[3]{2}$	$\sqrt[3]{3}$	$\sqrt[3]{4}$	$\sqrt[3]{5}$	$\sqrt[3]{6}$	$\sqrt[3]{7}$
1.260	1.442	1.587	1.710	1.817	1.913

Proposed by: Jefferson Zhou

Answer: $\frac{6}{325}$

Solution: Let $x = \sqrt[3]{a}$, and let $y = \sqrt[3]{b}$. Observe that the inequality rearranges to become $27 - a - b - 9\sqrt[3]{ab} = 3^3 + (-x)^3 + (-y)^3 - 3 \cdot 3(-x)(-y) = \frac{(3-x-y)((3+x)^2 + (3+y)^2 + (x-y)^2)}{2} \geq 0$. Also, since $3+x \neq 0$, the term $(3+x)^2 + (3+y)^2 + (x-y)^2$ is always positive, so the inequality simplifies to $3 - x - y \geq 0 \implies x + y \leq 3$.

We now do casework on a .

If $a = 0$, then clearly any $0 \leq b \leq 27$ works, yielding 28 solutions.

If $a = 1$, then any $0 \leq b \leq 8$ works, yielding 9 solutions.

If $a = 2$, then noticing that $\sqrt[3]{2} + \sqrt[3]{5} < 3$ but $\sqrt[3]{2} + \sqrt[3]{6} > 3$, any $0 \leq b \leq 5$ works, yielding 6 solutions.

If $a = 3$, then noticing that $\sqrt[3]{3} + \sqrt[3]{3} < 3$ but $\sqrt[3]{3} + \sqrt[3]{4} > 3$, any $0 \leq b \leq 3$ works, so there are 4 solutions.

Now, since $\sqrt[3]{4} + \sqrt[3]{4} > 3$, for any $4 \leq a \leq 27$, all solutions will just be a permutation of the solutions for $0 \leq a \leq 3$. From this, we deduce that for $4 \leq a \leq 5$, there are 3 solutions each, for any $6 \leq a \leq 8$, there are 2 solutions each, and for any $9 \leq a \leq 27$, there is only one solution, namely $(a, 0)$.

Now, the total number of solutions is $28 + 9 + 6 + 4 + 2 \cdot 3 + 3 \cdot 2 + 19 = 28 + 19 + 31 = 78$.

Since there are 65^2 pairs (a, b) , the answer is now $\frac{78}{65^2} = \frac{6}{5^2 \cdot 13} = \frac{6}{325}$.

Problem 21. [14] 9 boys and 4 girls are seated randomly around a circular table. Compute the probability that no two girls sit next to each other.

$$DE = 2 \cdot \frac{2}{2 + \sqrt{2} + \sqrt{6}} = \frac{4}{2 + \sqrt{2} + \sqrt{6}}.$$

$$\text{Now, } \frac{1}{\sqrt{4 - AE \cdot EB}} = \frac{1}{\sqrt{4 - (4 - DE^2)}} = \frac{1}{DE} = \boxed{\frac{2 + \sqrt{2} + \sqrt{6}}{4}}.$$

Problem 23. [17] Let $P(x)$ be the unique polynomial of minimal degree with coefficients in the set $\{0, 1, 2, 3, 4\}$ such that $(x^4 - x^3 + 2x^2 + 2x - 1)P(x)$ leaves the same remainder as x when divided by 5 for all positive integers x . Compute $P(10)$.

Proposed by: Jefferson Zhou

Answer: $\boxed{24100}$

Solution: Let $Q(x) = x^4 - x^3 + 2x^2 + 2x - 1$. We will work modulo 5.

Observe that $Q(0) \equiv 4 \pmod{5}$, $Q(1) \equiv 3 \pmod{5}$, $Q(2) \equiv 4 \pmod{5}$, $Q(3) \equiv 2 \pmod{5}$, $Q(4) \equiv 1 \pmod{5}$. It is necessary and sufficient for P to satisfy $P(x) \equiv x \cdot Q(x)^{-1} \pmod{5}$ for $x \in \{0, 1, 2, 3, 4\}$, so we must have $P(0) \equiv 0 \pmod{5}$, $P(1) \equiv 2 \pmod{5}$, $P(2) \equiv 3 \pmod{5}$, $P(3) \equiv 4 \pmod{5}$, and $P(4) \equiv 4 \pmod{5}$.

We now use finite differences. The finite difference table is shown below:

0	2	3	4	4
2	1	1	0	
-1	0	-1		
1	-1			
-2				

Denote the n th row from the bottom to be r_n .

Since $x^5 - x \equiv 0 \pmod{5}$ for all integers x and P is of minimal degree, $P(x)$ must be of degree at most 4 (or else the remainder when P is divided by $x^5 - x$ would also work, and would have a smaller degree). Thus, r_1 is a constant, and r_2 is linear with slope $-2 \equiv 3 \pmod{5}$, and so r_2 must be the function $3x + 1$.

Note that r_3 is quadratic, so let it be the degree 2 polynomial $f_3(x) = a_3x^2 + b_3x + c_3$. Row r_2 represents the function $f_3(x+1) - f_3(x) = a_3(2x+1) + b_3 \equiv 3x + 1 \pmod{5}$, so $a_3 = 3 \cdot 2^{-1} \equiv 4 \pmod{5}$, $b_3 \equiv 1 - a_3 \equiv 2 \pmod{5}$, and $c_3 \equiv f_3(0) \equiv -1 \pmod{5}$, and $f_3(x) = 4x^2 + 2x - 1$.

Now, r_4 is cubic, so let it be the degree 3 polynomial $f_4(x) = a_4x^3 + b_4x^2 + c_4x + d_4$. Row r_3 represents the function $f_4(x+1) - f_4(x) = a_4(3x^2 + 3x + 1) + b_4(2x + 1) + c_4 = 4x^2 + 2x - 1$. We have $a_4 \equiv 4 \cdot 3^{-1} \equiv 3 \pmod{5}$, $b_4 \equiv (2 - 3a_4) \cdot 2^{-1} \equiv -1 \pmod{5}$, $c_4 \equiv -1 - a_4 - b_4 \equiv 2 \pmod{5}$, and $d_4 = f_4(0) = 2$, so $f_4(x) = 3x^3 - x^2 + 2x + 2$.

Now, r_5 is quartic, and also happens to be $P(x)$, so let it be $ax^4 + bx^3 + cx^2 + dx + e$. Then, row r_4 represents the function $P(x+1) - P(x) \equiv a(4x^3 + x^2 + 4x + 1) + b(3x^2 + 3x + 1) + c(2x + 1) + d \equiv 3x^3 - x^2 + 2x + 2 \pmod{5}$, so $a \equiv 3 \cdot 4^{-1} \equiv 2 \pmod{5}$, $b \equiv (-1 - a)3^{-1} \equiv 4 \pmod{5}$, $c \equiv (2 - 3b - 4a)2^{-1} \equiv 1 \pmod{5}$, $d \equiv 2 - a - b - c \equiv 0 \pmod{5}$, and $e = P(0) = 0$, so $P(x) = 2x^4 + 4x^3 + x^2$. Now, $P(10) = \boxed{24100}$.

Note: It is also possible to do Lagrange Interpolation for this problem.

Problem 24. [17] A right circular cone has slant height 1. Compute the maximum possible volume of a rectangular prism inscribed within the cone.

Proposed by: Jefferson Zhou

Answer: $\boxed{\frac{16\sqrt{3}}{243}}$

Solution: Let b and a be the radius and height of the cone, respectively. We first find the maximum possible volume when b and a are fixed.

Say that the maximum possible volume occurs when the top face of the prism lies on a plane that cuts off a similar cone of radius r . Then, the height of this cone is $\frac{ar}{b}$, so the volume of the prism is the area of the top face multiplied by $a - \frac{ar}{b}$. The top face of the prism is inscribed in a circle of radius r , so its maximum area is well known to occur when it is a square, in which case it will have area $2r^2$.

Now, the volume of the prism is $2r^2 \left(a - \frac{ar}{b}\right) = \frac{2a}{b} \cdot r^2(b - r)$. We wish to maximize this, so we apply AM-GM; note that $\frac{2a}{b} \cdot r^2(b - r) = \frac{a}{b} \cdot r \cdot r \cdot (2b - 2r)$, and since $0 \leq r \leq b$, by AM-GM, $\frac{2b}{3} = \frac{r + r + 2b - 2r}{3} \geq \sqrt[3]{r \cdot r \cdot (2b - 2r)}$, so $r^2(2b - 2r) \leq \frac{8b^3}{27}$, with equality at $r = 2b - 2r$ so $r = \frac{2b}{3}$. Now, the maximum volume of the prism is $\frac{8b^3}{27} \cdot \frac{a}{b} = \frac{8ab^2}{27}$.

We now maximize this expression over all a, b . Since the slant height of the cone is 1, we have that $a^2 + b^2 = 1$, so $\frac{8ab^2}{27} = \frac{8a(1 - a^2)}{27} = \frac{8a(1 - a)(1 + a)}{27}$. We wish to use AM-GM again. As such, we want to multiply the terms $1 - a$ and $1 + a$ by constants z and y such that $(z - za) + (y + ya) + a$ is a constant, so $z = y + 1$. Also, we require that $(y + 1) - (y + 1)a = z - za = y + ya = a$ has a solution to find the maximum. Solving the equation, we yield $a = \frac{y}{1 - y}$ and $a = \frac{y + 1}{y + 2}$, so $\frac{y}{1 - y} = \frac{y + 1}{y + 2}$, so $1 - y^2 = y^2 + 2y$, and $2y^2 + 2y - 1 = 0$. By the quadratic formula, $y = \frac{-2 \pm \sqrt{12}}{4} = \frac{-2 \pm 2\sqrt{3}}{4} = \frac{-1 \pm \sqrt{3}}{2}$, but we wish for y to be positive, so $y = \frac{\sqrt{3} - 1}{2}$. Now, by AM-GM, $\frac{\sqrt{3}}{3} = \frac{2y + 1}{3} = \frac{(y + 1) - (y + 1)a + y + ya + a}{3} \geq \sqrt[3]{((y + 1) - (y + 1)a)(y + ya)a} = \sqrt[3]{(y + 1)y(1 - a)(1 + a)a} = \sqrt[3]{(y^2 + y)(1 - a)(1 + a)a} = \sqrt[3]{\frac{1}{2}(1 - a)(1 + a)a}$. Now, $(1 - a)(1 + a)a \leq 2 \cdot \left(\frac{\sqrt{3}}{9}\right) = \frac{2\sqrt{3}}{9}$, with equality at $a = \frac{y}{1 - y} = \frac{1}{\sqrt{3}}$. (We can easily test that $a(1 - a)(1 + a) = \frac{2\sqrt{3}}{9}$ at $a = \frac{1}{\sqrt{3}}$.)

Now, the maximum possible value of $\frac{8a(1 - a)(1 + a)}{27}$ is $\frac{8}{27} \cdot \frac{2\sqrt{3}}{9} = \boxed{\frac{16\sqrt{3}}{243}}$.

Problem 25. [20] Let p be the answer to problem 27. Sam starts with one stick of length 1. He begins to play a game with himself, where he randomly chooses a stick he currently has and breaks it perfectly in half, then chooses a new stick (possibly one of the halves) to break too, and so on. Given he breaks p sticks in total, find the expected value of the sum of the reciprocals of the lengths of the sticks. In other words, if the sticks after p breaks are of lengths (not necessarily distinct) $s_1, s_2, \dots, s_k$ then find the expected value of $\frac{1}{s_1} + \frac{1}{s_2} + \dots + \frac{1}{s_k}$.

Proposed by: Samuel Kretzschmar

Answer: $\boxed{286}$

Solution: Call the sum of the reciprocals of the lengths the *score* of a given set of sticks. Let $E[S(p)]$ be the expected value of the score after p breaks. We note that breaking a stick with length s_i will replace it two sticks of length $s_i/2$; this increases the score by $2 \cdot (\frac{2}{s_i}) - \frac{1}{s_i} = \frac{3}{s_i}$. We see the expected score ($\frac{1}{s_i}$) of a random stick after $p-1$ breaks must be $\frac{E[S(p-1)]}{p}$ as there are p sticks after $p-1$ breaks, so we find that

$$E[S(p)] = E[S(p-1)] + \frac{3}{p}E[S(p-1)] = \left(\frac{p+3}{p}\right)E[S(p-1)].$$

Noting $E[S(0)] = 1$, we find the the formula

$$E[S(p)] = \left(\frac{p+3}{p}\right)\left(\frac{p+2}{p-1}\right)\left(\frac{p+1}{p-2}\right)\left(\frac{p}{p-3}\right)\cdots\left(\frac{4}{1}\right) = \frac{(p+3)(p+2)(p+1)}{3 \cdot 2 \cdot 1} = \binom{p+3}{3}.$$

Finding $p = 10$, the answer is $\binom{13}{3} = 286$.

Problem 26. [20] Let q be the answer to problem 25. Let r be a real number, and let S be the set of ordered pairs of positive real numbers (x, y) such that $\sqrt{x} + \sqrt{y} = r$. The minimum possible value of $\sqrt{x+49} + \sqrt{y+64}$ across all ordered pairs in S is $\sqrt{q}$. Compute r .

Proposed by: Evan Truong

Answer: $\boxed{\sqrt{61}}$

Solution:

Consider points $O = (0, 0)$, $P = (7, \sqrt{x})$, and $Q = (7 + 8, \sqrt{x} + \sqrt{y})$. We see that $OP = \sqrt{(\sqrt{x})^2 + 7^2} = \sqrt{x+49}$, $PQ = \sqrt{(\sqrt{y})^2 + 8^2} = \sqrt{y+64}$, and $OQ = \sqrt{(\sqrt{x} + \sqrt{y})^2 + 15^2} = \sqrt{r^2 + 15^2}$. By the Triangle Inequality, $OP + PQ \geq OQ$. This equality is achievable for correctly chosen x, y , so the minimum possible value for $\sqrt{x+49} + \sqrt{y+64}$ is $\sqrt{q} = \sqrt{r^2 + 15^2}$. As $q = 286$, we have $r = \sqrt{q - 15^2} = \sqrt{286 - 225} = \sqrt{61}$.

Problem 27. [20] Let r be the answer to problem 26. In triangle ABC with $AC > AB$, let ω be the circumcircle. Point D lies on ω such that $AD \perp BC$. Let ℓ_1 be the line AD , ℓ_2 be the line BD , and let ℓ_3 be the reflection of ℓ_2 across ℓ_1 . If the radius of ω is r , and ℓ_3 intersects ω at a point $E \neq D$ such that $CE = 12$, what is the length of AD ?

Proposed by: Jefferson Zhou

Answer: $\boxed{10}$

Solution: The key is to prove that $AC \perp DE$. Say $X = AC \cap DE$ (intersection of AC and DE) and $F = AD \cap BC$. Then $\angle ADX = \angle ADB = \angle FDB = 90^\circ - \angle FBD = 90^\circ - \angle CBD = 90^\circ - \angle CAD = 90^\circ - \angle XAD$ so $\angle DXA = 90^\circ$, implying $AC \perp DE$. From this, we know $AD^2 + CE^2 = (2r)^2$; you can see this by considering triangle LMN inscribed in the circle with $LM = AD$ and $MN = CE$. Let O be the center of the circle. Since $\angle LON = \angle LOM + \angle MON = \angle AOD + \angle COD = 2 \cdot \angle AXD = 180^\circ$ by intersecting chord angles, LN is a diameter, $\angle LMN$ is right, and $AD^2 + CE^2 = LM^2 + MN^2 = LN^2 = (2r)^2$ as desired. Since $r = \sqrt{61}$, the answer is $AD = \sqrt{4 \cdot 61 - 12^2} = \sqrt{244 - 144} = 10$.

Note: To solve the cycle, observe that since $q = \binom{p+3}{3}$, we must have $r = \sqrt{q - 15^2}$, so $p^2 = (2r)^2 - CE^2 = 4(q - 15^2) - 144 = 4\binom{p+3}{3} - 1044$, so $1044 = 4\binom{p+3}{3} - p^2$. We know that p must be an integer, so we can just guess integers of reasonable size until we yield that $p = 10$, and then we can compute the other answers as above. (Note: the above equation is a cubic, so it can be verified that there are no other integer solutions.)